

RADAR

Real-Time Vehicle Analysis System

Egyptian License Plate Detection · OCR · Violation Detection · 2026

AI-Driven Traffic Enforcement Solutions

OVERVIEW 01

- Multi-stage pipeline for Egyptian traffic enforcement
- License plate detection, OCR, and super-resolution
- Vehicle detection with ByteTrack multi-object tracking
- Seatbelt and mobile phone violation detection
- Three parallel OCR models with confidence fusion
- AI-powered image enhancement for low-quality inputs

DETECTION PIPELINE 02



TECHNOLOGY STACK 03

YOLOv11m Plate Detection V1	ByteTrack Multi-Object Tracking
YOLO26m V2 Character OCR	LapSRN x2 Super-Resolution
YOLO26m V2-W3 Weighted OCR V3	Rea1-ESRGAN AI Enhancement
YOLO26s Vehicle Detection	YOLOv11m Seatbelt & Mobile

TRAINING RESULTS 04



KEY METRICS 05

3 OCR Models	94% Plate mAP	91% OCR Accuracy	25fps Detection Speed
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KEY CAPABILITIES 06

- Tri-Model OCR**
Three parallel OCR models with confidence-weighted fusion for maximum plate reading accuracy across varied conditions
- Super Resolution**
AI-powered image enhancement using LapSRN and Rea1-ESRGAN before OCR on low-quality or distant inputs
- Vehicle Tracking**
ByteTrack multi-object tracking with persistent identity across frames for real-time traffic monitoring
- Violation Detection**
Real-time seatbelt and mobile phone usage detection for automated traffic law enforcement